

SHELBY-CLEVELAND COUNTY AP, NC, USA

WMO: 720277

Lat: 35.256N

Lon: 81.601W

Elev: 258

StdP: 98.26

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63843

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.0	-4.6	-16.9	0.9	-1.9	-14.4	1.1	-0.4	10.2	7.0	8.6	7.7	1.5	30	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	34.0	22.9	32.7	22.6	31.9	22.4	24.8	30.0	24.3	29.4	23.9	28.8	2.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.8	26.0	22.9	18.2	25.8	22.5	17.7	25.6	76.6	29.7	74.6	29.7	72.6	28.8	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.7	5.4	-10.6	35.7	3.7	1.8	-13.2	37.0	-15.4	38.1	-17.4	39.2	-20.1	40.5		
(5)			-11.5	25.6	3.3	0.7	-13.8	26.2	-15.8	26.6	-17.6	27.0	-20.0	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.7	7.1	10.8	15.8	20.2	24.5	25.7	25.0	22.2	16.0	9.9	7.0		
	DBStd	8.40	5.29	5.14	5.04	4.06	3.33	2.34	2.00	2.39	2.96	4.32	4.34	4.92		
	HDD10.0	525	178	109	53	6	0	0	0	0	6	53	121			
	HDD18.3	1801	422	316	237	95	22	0	0	0	6	97	254	351		
	CDD10.0	2635	13	27	77	179	316	435	486	366	191	51	28			
	CDD18.3	871	0	1	4	19	79	185	228	208	121	24	1	1		
	CDH23.3	7647	0	4	43	223	670	1691	2071	1786	947	198	9	3		
	CDH26.7	2871	0	0	3	36	173	686	888	723	323	39	1	0		
(14) Wind	WSAvg	1.9	2.2	2.4	2.5	2.5	2.0	1.6	1.5	1.3	1.7	1.8	1.8	1.9		
(15) Precipitation	PrecAvg	1226	108	89	126	103	88	114	106	115	99	91	85	103		
	PrecMax	1735	191	190	220	257	219	269	260	198	298	307	212	202		
	PrecMin	708	26	23	33	29	22	23	35	16	3	0	27	49		
	PrecStd	251	51	39	48	57	47	64	47	56	60	74	46	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	23.6	27.2	29.8	32.3	35.3	35.5	36.3	33.1	30.7	24.9	22.7		
		MCWB	15.5	16.6	15.9	17.4	20.5	22.2	22.9	22.1	21.4	21.5	17.0	16.3		
	2%	DB	17.8	20.7	24.5	27.6	30.2	33.3	33.7	33.9	32.0	27.4	22.2	19.8		
		MCWB	13.6	14.4	15.1	16.9	19.9	22.3	23.3	23.2	21.3	19.2	15.0	15.8		
	5%	DB	15.9	18.1	22.2	26.1	28.6	32.2	32.6	32.3	30.2	25.6	20.1	17.5		
		MCWB	11.7	12.9	14.0	16.6	20.1	22.2	23.1	22.9	21.4	18.7	14.9	14.1		
	10%	DB	13.0	16.2	19.9	24.1	27.2	30.7	31.3	30.6	28.4	23.4	17.8	15.2		
		MCWB	9.8	11.2	13.0	15.8	19.1	21.9	22.9	22.9	21.2	18.1	13.6	12.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.1	17.8	17.8	20.7	23.0	25.0	25.4	25.4	24.1	23.4	20.0	19.1		
		MCDB	18.6	21.6	22.4	24.1	27.3	29.9	30.9	30.5	28.9	26.4	21.3	20.3		
	2%	WB	15.2	16.3	16.8	19.4	21.9	24.1	24.9	24.8	23.5	21.9	17.8	17.1		
		MCDB	17.4	18.9	20.7	24.0	26.1	29.6	30.4	29.7	28.0	24.7	19.5	18.9		
	5%	WB	12.9	14.6	15.7	18.2	21.4	23.6	24.4	24.3	22.8	20.6	16.4	15.2		
		MCDB	14.7	17.4	20.0	23.0	25.6	29.0	29.7	29.1	27.1	23.2	18.4	16.7		
	10%	WB	10.5	12.1	14.4	17.3	20.6	23.0	23.9	23.7	22.2	19.4	14.7	12.6		
		MCDB	12.7	14.8	18.9	21.8	24.9	28.2	29.0	28.3	26.2	22.6	17.5	14.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	11.1	11.9	12.5	11.5	11.2	10.6	10.3	10.6	11.8	12.2	10.1		
		MCDBR	13.3	14.0	14.4	15.1	13.4	13.0	12.3	12.4	13.2	13.6	14.6	12.2		
	5% WB	MCWBR	9.1	8.7	7.1	6.5	5.1	4.1	3.6	3.5	4.5	6.4	8.5	8.1		
		MCWBR	10.5	11.9	11.7	11.9	10.3	11.0	10.5	10.2	10.4	10.0	10.5	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.317	0.352	0.399	0.454	0.482	0.528	0.516	0.421	0.373	0.328	0.310		
	taud		2.511	2.480	2.418	2.281	2.162	2.122	2.020	2.034	2.286	2.403	2.490	2.536		
	Ebn at Noon		898	929	918	886	838	811	771	771	836	851	866	874		
	Edh at Noon		82	94	110	132	151	157	173	167	124	101	83	76		
(44) All-Sky Solar Radiation	RadAvg		2.64	3.29	4.33	5.39	5.99	6.34	5.92	5.46	4.66	3.85	2.92	2.26		
	RadStd		0.15	0.37	0.30	0.42	0.48	0.45	0.41	0.26	0.42	0.46	0.25	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+108

Nomenclature: See separate page